```json
{
  "product_name": "Google AI Bridge",
  "product_vision": "To transform any local AI system into a super-enhanced, high-intelligence, high-resource model by providing a unified, intelligent, and secure local bridge that seamlessly connects it to the entire spectrum of Google's AI and data platform APIs. Our vision is to democratize access to Google's cutting-edge capabilities, allowing local AI to transcend its computational limits and leverage the vast potential of the cloud for unprecedented functionality, consistency, and contextual awareness.",
  "target_audience": [
    "Local AI System Developers (home automation, robotics, personal assistants)",
    "Knowledge Workers (document management, design, research)",
    "Tech Enthusiasts & Hobbyists",
    "Businesses with On-Premise Data Needs"
  ],
  "core_problems_solved": [
    "Limited local AI intelligence & resources",
    "Fragmented API integration complexity",
    "Context drift & inconsistent AI output",
    "Balancing data privacy with cloud power",
    "Managing cloud API costs & efficiency"
  ],
  "solution_architecture_overview": {
    "description": "The Google AI Bridge operates as a sophisticated hybrid system. Its core intelligence resides in the Google Cloud (managed via Google AI Studio). The local application acts as the universal connector, routing requests, executing tool calls, and managing authentication for both a cloud-based AI Agent and an optional user-configured local LLM.",
    "components": [
      {
        "name": "Cloud-Based AI Agent (AI Studio)",
        "role": "Central intelligence, orchestrator, decision-maker for cloud operations",
        "llm": "Google Gemini (Pro, Pro Vision)",
        "platform": "Google AI Studio"
      },
      {
        "name": "Local Google AI Bridge Application (The Universal Connector)",
        "role": "Unified API endpoint, intelligent router, secure authenticator, tool executor",
        "platform": "Cross-platform desktop (Windows, macOS, Linux)"
      },
      {
        "name": "Optional User Localized LLM",
        "role": "User-configured local intelligence for privacy/latency-sensitive tasks, leverages Bridge as a tool provider",
        "example": "Ollama hosting Mistral/Qwen"
      }
    ]
  },
  "development_plan": {
    "methodology": "Agile, iterative development with clear phase deliverables.",
    "core_tooling": [
      "Backend: Go (for performance, concurrency, single binary)",
      "Frontend: Electron (for cross-platform desktop UI), React/TypeScript (for UI components), Material UI (for Material Design 3 adherence)",
      "Database: SQLite (for local configuration, encrypted credentials)",
      "Container Runtime (for optional local LLM integration): `containerd` + `nerdctl` (or similar lean, embedded solution)",
      "Version Control: Git (GitHub)",
      "CI/CD: GitHub Actions",
      "Diagramming: PlantUML or Mermaid (within Markdown)"
    ],
    "phases": [
      {
        "name": "Phase 1: Core Connectivity & AI Studio Agent Foundation (4-6 months)",
        "goals": [
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        "Establish robust communication between local app and Google AI Studio Agent.",
        "Implement core authentication and a foundational set of Google API tools for Gemini."
      ],
      "key_features": [
        "**Local AI Bridge Backend (Go):**",
        "- RESTful HTTP API server (`localhost:8080`).",
        "- Client for Google AI Studio Gemini API.",
        "- Secure credential storage (encrypted SQLite).",
        "- Initial Tool Execution Engine for core Google APIs.",
        "- Basic Intelligent Router (direct to AI Studio by default).",
        "**Local AI Bridge UI (Electron/React):**",
        "- Dashboard (connection status, basic logs).",
        "- Settings: AI Studio API Key input, Google Account OAuth setup (basic: Gmail, Drive, Calendar), 'Agent Brain' copy/paste for System Prompt & Tool JSON.",
        "- **Chat Interface (Default):** Unified chat for interacting with the AI Studio Agent. Visual indicators for processing and tool calls.",
        "**Cloud-Based AI Agent (Google AI Studio - Gemini):**",
        "- Core System Prompt (orchestrator persona).",
        "- Initial Tool Definitions (schemas for `gmail_search_emails`, `drive_search_files`, `google_custom_search`, `vertex_ai_generative_ai_text_summarize`).",
        "- Few-Shot Examples for basic multi-tool orchestration."
      ],
      "ui_elements_focus": [
        "Dashboard: 'AI Studio Connection Status' (●/●), 'Google Accounts Connected' (icons for Gmail, Drive, Calendar).",
        "Settings Tab:",
        "  - 'AI Studio API Key': Text input, 'Test Connection' button.",
        "  - 'Google Account Connections': Buttons for each (Gmail, Drive, Calendar), displaying 'Connected' status & scopes.",
        "  - 'Agent Brain (from AI Studio)':",
        "    - Text Area: 'System Prompt' (placeholder with instructions for copy/paste from AI Studio).",
        "    - Text Area: 'Tool Definitions (JSON)' (placeholder with instructions for copy/paste from AI Studio).",
        "  - 'Local LLM Integration': (Placeholder for Phase 2).",
        "Chat Interface Tab:",
        "  - Main text input, send button.",
        "  - Conversation history display.",
        "  - Status indicators (e.g., 'Processing...', 'Calling Gmail API...')."
      ],
      "ai_studio_requirements": [
        "Create a new AI Studio project for the agent.",
        "Enable Gemini API.",
        "Define a robust, orchestrator-focused System Prompt.",
        "Define Tool Definitions for: Gmail API (search/read), Drive API (search/read), Google Custom Search API, Vertex AI Generative AI (text summarization, content generation).",
        "Curate initial Few-Shot Examples demonstrating multi-step tool use and synthesis."
      ]
    },
    {
      "name": "Phase 2: Comprehensive API Integration & Local LLM Hook (6-9 months)",
      "goals": [
        "Integrate all specified Google APIs as tools for the AI Studio Agent.",
        "Provide a clear interface for an optional user-configured local LLM.",
        "Refine intelligent routing."
      ],
      "key_features": [
        "**Local AI Bridge Backend (Go):**",
        "- Full implementation of all specified Google APIs as callable tools.",
        "- Implementation of 'local' tools (e.g., `local_read_file`, `local_list_directory`).",
        "- Enhanced Intelligent Router with user-configurable rules (e.g., 'if
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        specific keyword, route to local LLM').",
        "- Local LLM Tool Provider: Expose Google API tool schemas to external local LLMs.",
        "- API Usage & Cost Monitoring.",
        "**Local AI Bridge UI (Electron/React):**",
        "- Settings: Dedicated sections for each Google API (checkboxes to enable, text inputs for keys/IDs, clear OAuth status).",
        "- **Local LLM Integration settings:** Endpoint configuration (`http://localhost:XXXX`), 'Test Connection' button.",
        "- **Dedicated Local LLM Chat Tab:** A separate chat interface that routes prompts specifically to the configured local LLM (if enabled), and displays its responses and any tool calls it tries to make *back to the Bridge*.",
        "- Real-time API usage and estimated cost display.",
        "**Cloud-Based AI Agent (Google AI Studio - Gemini):**",
        "- Define Tool Definitions for *all* remaining specified Google APIs (Cloud Vision, Speech, NLP, Translation, Document AI, Video Intelligence, Places, People, Contacts, Blogger, Workspace Add-ons, Slides, Apps Script, Sheets, Groups Settings, Drive Activity, Meet, Vault, Forms, Keep, Workspace Events, Docs, Calendar, Chat, Tasks, Fact Check Tools).",
        "- Refine System Prompt for complex multi-API orchestration and edge cases.",
        "- Expand Few-Shot Examples for all new API integrations and cross-API workflows."
      ],
      "ui_elements_focus": [
        "Settings Tab:",
        "  - 'Google AI/ML APIs': Grouped checkboxes & input fields for specific keys/project IDs for: Gemini API, Document AI Warehouse, Gemini Cloud Assist, Sensitive Data Protection (DLP), Gemini for Google Cloud, AI Platform Training & Prediction API, Cloud AutoML API, Cloud Natural Language API, Cloud Optimization API, Cloud Speech-to-Text API, Cloud Translation API, Cloud Video Intelligence API, Cloud Vision API, Dialogflow API, Vertex AI API, Places API.",
        "  - 'Google Workspace & Other APIs': Grouped checkboxes & input fields/OAuth buttons for: YouTube Data API v3, Google People API, Contacts API, Blogger API, Google Workspace Add-ons API, Google Slides API, Apps Script API, Google Sheets API, Groups Settings API, Google Drive Activity API, Google Meet REST API, Google Vault API, Google Forms API, Google Keep API, Google Workspace Events API, Google Docs API, Google Calendar API, Google Chat API, Google Tasks API, Gmail API, Google Drive API, Fact Check Tools API.",
        "  - 'Local LLM Integration':",
        "    - Input: 'Local LLM Endpoint URL' (e.g., `http://localhost:11434`).",
        "    - Button: 'Test Connection'.",
        "    - Checkbox: 'Enable Local LLM as primary route'.",
        "New Chat Tab: 'Local LLM Chat'",
        "  - Appears only if 'Local LLM Endpoint URL' is configured and connection is successful.",
        "  - Dedicated chat interface to send prompts to the local LLM. Responses display normally.",
        "  - Visual debug for tool calls the *local LLM* attempts to make to the Bridge."
      ],
      "ai_studio_requirements": [
        "Define Tool Definitions for *all* listed Google APIs. This is a massive effort requiring precise `name`, `description`, `input_schema` (parameters), and expected `output_schema` for each. This forms the contract for the AI Studio Agent's understanding and the Bridge's execution.",
        "Refine System Prompt to handle the vast array of new tools, prioritizing their use, and synthesizing complex information from multiple sources.",
        "Add extensive Few-Shot Examples demonstrating cross-API workflows (e.g., 'Summarize meeting, then find related emails, then draft a reply')."
      ]
    },
    {
      "name": "Phase 3: Refinement, Performance & Enterprise Features (6-12+ months)",
      "goals": [
        "Optimize performance and stability across all components.",
        "Enhance UI/UX based on user feedback.",
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            "Explore advanced features for enterprise users (e.g., DLP integration features)."
          ],
          "key_features": [
            "Automated updates for the Local Google AI Bridge application.",
            "Advanced logging & debugging tools (e.g., request tracing, cost breakdowns per query).",
            "Performance optimizations for API calls (caching, parallel execution).",
            "Granular access control for Google API scopes (per AI agent profile).",
            "DLP integration features (e.g., pre-flight scanning of data before sending to cloud, or analysis of local files for sensitive data using DLP API).",
            "Enhanced routing rules (e.g., context-aware, user-defined thresholds).",
            "Community contribution pathways (for open-source core)."
          ],
          "ui_elements_focus": [
            "Dashboard: Detailed cost breakdowns, API call volume graphs.",
            "Settings: Granular permissions for each API access, DLP rules configuration.",
            "Chat: Improved context management visuals, AI explanations for reasoning steps."
          ],
          "ai_studio_requirements": [
            "Continual refinement of System Prompt and Tool Definitions based on real-world usage data and edge cases.",
            "Exploration of Vertex AI Model Garden for specialized LLM tasks if Gemini proves insufficient in specific domains.",
            "Integration with Vertex AI Vector Search for large-scale RAG on user's cloud-stored data (if user opts-in to billable service)."
          ]
        }
      ]
    },
    "local_llm_integration_details": {
      "communication_protocol": "HTTP REST API (expected from local LLM, e.g., Ollama's API)",
      "interface_exposed_by_local_llm": "Expected to support tool-calling functionality, similar to how Gemini handles it, by generating structured JSON tool calls.",
      "user_configuration": "Yes, endpoint URL (e.g., `http://localhost:11434`), optional API key/authentication details.",
      "dedicated_chat_function": "A dedicated 'Local LLM Chat' tab will appear in the UI once the local LLM endpoint is successfully configured and connected. This chat tab will exclusively send prompts to the user's local LLM and display its responses. If the local LLM attempts to make a tool call (that the Bridge exposes), the Bridge will execute it and return the result to the local LLM for its final response formulation."
    },
    "google_api_interaction_details": {
      "authorization_flow": {
        "oauth2": "For Google Workspace (Gmail, Drive, Calendar, People, Contacts, etc.), YouTube Data API, Google Maps for Fleet Routing, Places API: Bridge initiates OAuth 2.0 flow via user's browser, user grants consent, Bridge securely stores refresh tokens.",
        "api_keys_service_accounts": "For Google AI Studio Endpoint, Google Cloud ML APIs (Vision, Speech, Document AI, etc.), Google Custom Search API: User generates API Key or downloads Service Account JSON from Google Cloud Console and inputs/uploads to Bridge UI. Bridge stores credentials securely."
      },
      "llm_api_interaction_mechanism": "Function Calling (aka Tool Calling). Both the Cloud-Based AI Agent (Gemini) and the optionally configured Local LLM will be supplied with the schemas of available Google API tools. They will then generate structured JSON `function_call` requests. The Local Google AI Bridge's backend will execute these calls, retrieve results, and feed them back to the respective LLM for final response synthesis."
    },
    "ui_ux_flow_details": {
      "main_interface_look": "Central Chat Interface, with a sidebar/tabs for Dashboard and Settings.",
      "chat_interface_design": "Clean, Material Design 3 adherence. Text input at bottom.
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      Conversation history above. Visual cues for AI processing (spinners, subtle animations). Icons to indicate API usage (e.g., small Gmail icon next to an email summary). Clickable elements for rich media outputs (e.g., 'View OCR Result', 'Listen to Audio Output').",
      "dashboard_design": "Visualized API usage metrics (graphs for calls/cost per API), connection status to AI Studio and various Google services, recent activity log with filterable events.",
      "settings_design": "Organized into logical sections (e.g., 'AI Agent Brain', 'Google Account Integrations', 'AI/ML APIs', 'Workspace & Other APIs', 'Local LLM Integration', 'Routing Rules'). Each API/integration will have a clear enable/disable toggle, status indicator, and configuration fields (API key input, OAuth button, project ID if applicable).",
      "typical_user_scenario_example": {
        "scenario": "User types: 'Summarize my calendar events for tomorrow, then find urgent emails from 'client@example.com' about our new project, and draft a short reply asking for more details.'",
        "flow": [
          "**1. User Input:** User types prompt into Google AI Bridge chat.",
          "**2. Bridge Routing:** Bridge receives prompt. Its Intelligent Router (configured to prioritize the Cloud-Based AI Agent for complex tasks) sends the prompt + conversation history to the AI Studio Agent's endpoint.",
          "**3. AI Studio Agent Reasoning:** AI Studio Agent (Gemini) receives prompt. It determines based on its internal logic, prompt, and tool definitions that it needs:",
          "    a. `calendar_get_events(start_time='tomorrow_start', end_time='tomorrow_end')`",
          "    b. `gmail_search_emails(query='urgent new project', sender='client@example.com')`",
          "    c. `gmail_get_email_content(message_id='<from_search_result>')`",
          "    d. `gmail_draft_email(to='client@example.com', subject='Re: New Project', body='...')`",
          "    AI Studio Agent sends a structured JSON response containing these `function_call` requests back to the Bridge.",
          "**4. Bridge Executes API Calls & Authentication:** Bridge receives `function_call` requests. For each, it:",
          "    a. Uses securely stored OAuth 2.0 tokens for Calendar and Gmail APIs.",
          "    b. Makes the actual Google API calls.",
          "    c. Handles all API responses, rate limits, and error conditions.",
          "**5. Bridge Returns Results to AI Studio Agent:** Bridge packages results from Calendar and Gmail API calls (e.g., event summaries, email content) into a structured JSON payload and sends it back to the AI Studio Agent as a 'tool response' within the ongoing conversation context.",
          "**6. AI Studio Agent Synthesizes & Drafts:** AI Studio Agent receives the API results. It synthesizes calendar info and email content. It then uses its internal logic and the `gmail_draft_email` tool to generate the email body based on the context.",
          "**7. AI Studio Agent Sends Final Response:** AI Studio Agent sends its final, synthesized text response (e.g., 'Here's your summary of tomorrow's meetings and urgent emails. I've drafted a reply for the client: [Drafted Email Content]') back to the Bridge.",
          "**8. Bridge Displays to User:** Bridge displays the comprehensive response in its chat interface, including visual cues for API usage and estimated costs."
        ]
      }
    }
}
```